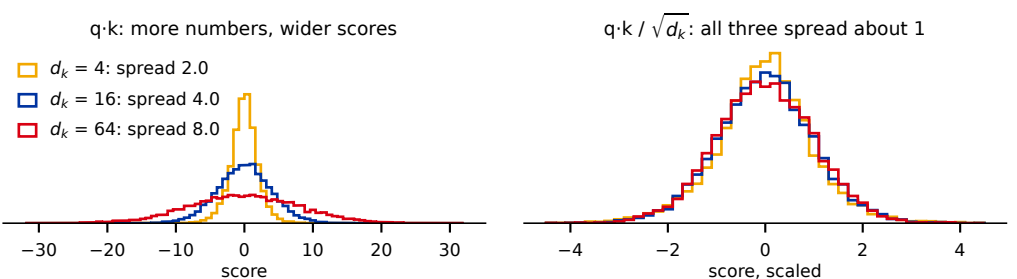
q·k: more numbers, wider scores
d_k = 4: spread 2.0
d_k = 16: spread 4.0
d_k = 64: spread 8.0
−30
−20
−10
0
10
20
30
score
q·k / $\sqrt{d_k}$: all three spread about 1
−4
−2
0
2
4
score, scaled